Troy Wu

troywu5@gmail.com | linkedin.com/in/troy-wu | github.com/troy-wu | troy-wu.github.io

Education

University of British Columbia Expected Graduation: November 2027
Bachelor of Computer Science and Business | ***GPA: 3.85/4.00*** *Vancouver, BC*

- Dean's Honour Roll, Trek Excellence Scholarship (Top 5%), Ranked 6/791 in UBC Sauder.
- **Courses:** Data Structures & Algorithms; OOP (Java); Operating Systems; Relational Databases.

Experience

Tesla August 2026 - December 2026
Software Engineer Intern - Maps and Navigation *Palo Alto, CA*

- Built an AI agent that matches social media navigation complaints to production telemetry trips at **95% precision**.
- Engineered the agent's retrieval layer to rank candidates across **100B+ rows** of fleet navigation telemetry per query, combining LLM vision geocoded map labels with parallel Trino queries.
- Productionized the agent as an event-driven service fed by a Jenkins scraping pipeline, posting high confidence matches to Teams automatically at **40 cents per post**.

Amazon May 2026 - August 2026
Software Development Engineer Intern - Everyday Essentials *Vancouver, BC*

- Served as the **first engineer in a team of applied scientists**, automating and productionizing machine learning infrastructure from research prototype to scaled production pipeline.
- Designed and built an end-to-end Python ML inference pipeline scoring product relationships across **15B+ product pairs**, powering recommendations attributed to **$330M+ annualized sales** across Amazon NA storefronts.
- Reduced batch inference time from **4 days to 8 hours** by orchestrating 80+ parallel GPU jobs on g5.4xlarge instances with **Airflow and AWS Batch**, with built-in fault tolerance and shard-level resumability.
- Designed a S3 feature store serving **1TB of product embeddings** via parallel batch lookups, eliminating a 90-minute fetch bottleneck and cutting per-run data costs from **$13 to under $1**.

D2L (Desire2Learn) January 2026 - April 2026
Software Engineer Intern - Lumi (Brightspace GenAI) *Vancouver, BC*

- Developed full-stack GenAI features for Brightspace LMS deployed across **400+ institutions** and **260K+ annual generations** using Web Components/LitElement, C# MVC, and SQL databases.
- Built Brightspace's **AI Audio Overview** feature end-to-end, delivering natural AI-narrated content summaries via serverless **Claude Sonnet + AWS Bedrock/Lambda and Polly TTS** pipeline to support diverse learning styles.
- Built and refactored Playwright E2E tests for critical GenAI workflows **expanding test coverage by 28%**, and consolidated duplicate code through shared TestBench fixtures **reducing suite runtime by 32%**.

Absolute Security May 2024 - December 2024
Software Engineer Intern *Vancouver, BC*

- Built and deployed full-stack, telemetry security services (Java, JavaScript/Manifest V3 extension) for 1M+ enterprise Chromebook endpoint devices, **enabling continuous monitoring and threat detection**.
- Engineered a scalable Java microservice integrated with **AWS Grafana** to automate and standardize log exports for 1M+ devices, **boosting error detection speed by 30%** and QA efficiency.
- Rebuilt QA log interface (HTML, CSS, JS), **reducing export latency by 50%** and improving usability.

Projects

BasketBot – ProduHacks Winner | *Python, OpenCV, TensorFlow, PyAudio, VOSK*

- Winner of ProduHacks (50 teams, $400) by building an object-detection-based automatic basketball scorekeeper.
- Implemented TensorFlow-based machine learning models to recognize basketball nets for accurate scoring.

Technical Skills

Languages: Python, Java, C#, TypeScript, JavaScript, C++, SQL.
Frameworks/Libraries: PyTorch, TensorFlow, pandas, NumPy, React.js, Node.js, Express.js, Flask, OpenCV, JUnit.
Developer Tools: AWS (Batch, DynamoDB, Bedrock, Lambda, S3), Airflow Docker, Kubernetes, Databricks, Git, MongoDB.